Lingofy : Full Database Schema

1. Users

Field	Type	Description
user_id	UUID (PK)	Unique user identifier
name	VARCHAR	User name
email	VARCHAR (unique)	Login email
password_hash	TEXT	Encrypted password
native_language	VARCHAR	User's first language
learning_language	VARCHAR	Language being learned
created_at	TIMESTAMP	Account creation
last_login	TIMESTAMP	Last activity
research_consent	BOOLEAN	Consent for research data
preferred_learning_path	ENUM(traditional, music)	

2. User Preferences

Stores music and learning preferences.

Field	Type
preference_id	UUID (PK)
user_id	UUID (FK)
favorite_genres	JSON
favorite_artists	JSON
vocabulary_level	ENUM (beginner/intermediate/advanced)
grammar_focus	BOOLEAN
phrase_learning	BOOLEAN
session_goal_minutes	INTEGER

3. Artists

Field	Type
artist_id	UUID (PK)
name	VARCHAR
genre	VARCHAR
country	VARCHAR

4. Albums

Field	Type
album_id	UUID (PK)
artist_id	UUID (FK)
title	VARCHAR
release_date	DATE
cover_image_url	TEXT

5. Songs

Field	Type
song_id	UUID (PK)
album_id	UUID (FK)
artist_id	UUID (FK)
title	VARCHAR
language	VARCHAR
duration_seconds	INTEGER
audio_url	TEXT
instrumental_url	TEXT

Field	Type
bpm	INTEGER
difficulty_level	ENUM(beginner, intermediate, advanced)

6. Lyrics

Field	Type
lyrics_id	UUID (PK)
song_id	UUID (FK)
full_lyrics	TEXT

7. Lyric Segments

Field	Type
segment_id	UUID (PK)
song_id	UUID (FK)
segment_order	INTEGER
text	TEXT
start_time	FLOAT
end_time	FLOAT

Field	Type
vocabulary_difficulty	ENUM
grammar_tag	VARCHAR

8. Vocabulary Extraction

Field	Type
word_id	UUID (PK)
segment_id	UUID (FK)
word	VARCHAR
translation	VARCHAR
part_of_speech	VARCHAR
phonetic	VARCHAR
difficulty_level	ENUM

9. Learning Paths

Field	Type
path_id	UUID (PK)
name	ENUM(traditional, music_integrated)
description	TEXT

10. User Experiment Assignment

Field	Type
assignment_id	UUID (PK)
user_id	UUID (FK)
path_id	UUID (FK)
assigned_at	TIMESTAMP

11. Learning Sessions

Field	Type
session_id	UUID (PK)
user_id	UUID (FK)
start_time	TIMESTAMP

Field	Type
end_time	TIMESTAMP
playback_mode	ENUM(text_only, instrumental, full_music)
tempo_multiplier	FLOAT
completion_status	BOOLEAN

12. Segment Interaction Logs

Field	Type
interaction_id	UUID (PK)
session_id	UUID (FK)
segment_id	UUID (FK)
time_spent_seconds	FLOAT
taps_count	INTEGER
reorder_attempts	INTEGER
fill_blank_attempts	INTEGER
mistakes	INTEGER

13. Error Tracking

Field	Type
error_id	UUID (PK)
interaction_id	UUID (FK)
word_id	UUID (FK)
error_type	ENUM(semantic, phonetic, spelling)
correction_time	FLOAT

14. Adaptive Tempo Tracking

Field	Type
tempo_event_id	UUID (PK)
session_id	UUID (FK)
segment_id	UUID (FK)
previous_tempo	FLOAT
new_tempo	FLOAT
trigger_reason	ENUM(low_accuracy, hesitation, smooth_progress)

15. Immediate Recall Quiz

Field	Type
quiz_id	UUID (PK)
session_id	UUID (FK)
segment_id	UUID (FK)
question_type	ENUM(fill_blank, translation, reorder)
user_answer	TEXT
correct_answer	TEXT
is_correct	BOOLEAN
response_time	FLOAT

16. Delayed Recall Quiz

Field	Type
delayed_quiz_id	UUID (PK)
user_id	UUID (FK)
segment_id	UUID (FK)
scheduled_time	TIMESTAMP
attempt_time	TIMESTAMP

Field	Type
score	FLOAT

17. Cognitive Load Self-Assessment

Field	Type
load_id	UUID (PK)
session_id	UUID (FK)
effort_score	INTEGER (1–10)
difficulty_rating	INTEGER
distraction_rating	INTEGER

18. Reflection Responses

Field	Type
reflection_id	UUID (PK)
session_id	UUID (FK)
helped_memory	TEXT
distractions	TEXT
music_effect	ENUM(helped, neutral, hindered)

Field	Type
additional_notes	TEXT

19. Engagement Analytics

Field	Type
analytics_id	UUID (PK)
user_id	UUID (FK)
session_id	UUID (FK)
voluntary_session_length	INTEGER
songs_skipped	INTEGER
segments_completed	INTEGER
retention_rate	FLOAT

20. Return Behavior Tracking

Field	Type
return_id	UUID (PK)
user_id	UUID (FK)
days_between_sessions	INTEGER

Field	Type
weekly_sessions	INTEGER

21. Song Recommendation Log

Field	Type
recommendation_id	UUID (PK)
user_id	UUID (FK)
song_id	UUID (FK)
recommendation_reason	VARCHAR
accepted	BOOLEAN
skipped	BOOLEAN

22. Playlists

Field	Type
playlist_id	UUID (PK)
user_id	UUID (FK)
title	VARCHAR

Field	Type
created_at	TIMESTAMP

23. Playlist Songs

Field	Type
id	UUID (PK)
playlist_id	UUID (FK)
song_id	UUID (FK)
added_at	TIMESTAMP

Lingofy ER Relationship Diagram

1. User Core Relationships

```
Users
  ├── UserPreferences
  ├── LearningSessions
  │       ├── SegmentInteractions
  │       │       ├── ErrorTracking
  │       │       └── TempoEvents
  │       │
  │       ├── ImmediateQuiz
  │       ├── CognitiveLoadAssessment
  │       └── ReflectionResponses
  │
  ├── DelayedRecallQuiz
  ├── EngagementAnalytics
  ├── ReturnBehavior
  ├── SongRecommendationLog
  └── Playlists
          └── PlaylistSongs
```

Relationships

```
Users (1) ──────── (1) UserPreferences

Users (1) ──────── (M) LearningSessions

Users (1) ──────── (M) DelayedRecallQuiz

Users (1) ──────── (M) Playlists

Users (1) ──────── (M) EngagementAnalytics

Users (1) ──────── (M) ReturnBehavior

Users (1) ──────── (M) SongRecommendationLog
```

2. Music Content Structure

```
Artists
   │
   └── Albums
         │
         └── Songs
               │
               ├── Lyrics
               │
               └── LyricSegments
                     │
                     └── Vocabulary
```

Relationships

```
Artists (1) ──────── (M) Albums

Artists (1) ──────── (M) Songs

Albums (1) ──────── (M) Songs

Songs (1) ──────── (1) Lyrics

Songs (1) ──────── (M) LyricSegments

LyricSegments (1) ──────── (M) Vocabulary
```

3. Learning Interaction Layer

Tracks cognition and learning behavior.

```
LearningSessions
      │
      ├── SegmentInteractions
      │        ├── ErrorTracking
      │        └── TempoEvents
      │
      ├── ImmediateQuiz
      ├── CognitiveLoadAssessment
      └── ReflectionResponses
```

Relationships

```
LearningSessions (1) ──────── (M) SegmentInteractions

SegmentInteractions (1) ──────── (M) ErrorTracking

SegmentInteractions (1) ──────── (M) TempoEvents

LearningSessions (1) ──────── (M) ImmediateQuiz

LearningSessions (1) ──────── (1) CognitiveLoadAssessment

LearningSessions (1) ──────── (1) ReflectionResponses
```

4. Experiment Design Layer

Supports control vs experimental groups.

```
Users
  |
  └── ExperimentAssignment
        |
        └── LearningPath
```

Relationships

```
Users (1) ──────── (1) ExperimentAssignment

LearningPath (1) ──────── (M) ExperimentAssignment
```

5. Song Interaction & Recommendation

```
Songs
  |
  ├── SongRecommendationLog
  |
  └── PlaylistSongs
        |
        └── Playlists
```

Relationships

```
Songs (1) ──────── (M) SongRecommendationLog

Songs (1) ──────── (M) PlaylistSongs

Playlists (1) ──────── (M) PlaylistSongs
```

6. Learning Interaction With Lyrics

Critical Lingofy learning system.

```
Songs
   │
   └── LyricSegments
           │
           ├── SegmentInteractions
           │
           ├── ImmediateQuiz
           │
           └── DelayedRecallQuiz
```

Relationships

```
LyricSegments (1) ──────── (M) SegmentInteractions

LyricSegments (1) ──────── (M) ImmediateQuiz

LyricSegments (1) ──────── (M) DelayedRecallQuiz
```

Complete ER Overview

```
Users
 ├── Preferences
 ├── ExperimentAssignment — LearningPath
 ├── LearningSessions
 │       ├── SegmentInteractions
 │       │       ├── Errors
 │       │       └── TempoEvents
 │       │
 │       ├── ImmediateQuiz
 │       ├── CognitiveLoad
 │       └── Reflection
 │
 ├── DelayedQuiz
 ├── EngagementAnalytics
 ├── Recommendations
 └── Playlists
        └── PlaylistSongs

Artists
 └── Albums
      └── Songs
           ├── Lyrics
           └── LyricSegments
                └── Vocabulary
```

Key Relationship Types

Entity A	Relationship	Entity B	Cardinality
User	→ Sessions	Sessions	1 : Many
Song	→ LyricSegments	LyricSegments	1 : Many
Segment	→ Vocabulary	Vocabulary	1 : Many
Session	→ SegmentInteractions	SegmentInteractions	1 : Many
Interaction	→ Errors	Errors	1 : Many
Users	→ Playlists	Playlists	1 : Many
Playlists	↔ Songs	Songs	Many : Many
Users	→ ExperimentAssignment	ExperimentAssignment	1 : 1